

# **LOG BOOK**

## **FOR**

### **“Virtual Herbal Garden”**

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## Log Book

The log book (**GitHub content**) includes the following contents. The details should be included with appropriate split up week wise for **Project Stage-I (Sem-VI, 2024-25)**.

| Sr. No. | Contents                                                                                                                                                                                                                     | Weekly Date              |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 1       | Introduction<br>-Problem Statement<br>-Objectives<br>-Application                                                                                                                                                            | 24/02/2025 to 08/03/2025 |
| 2       | Literature Survey<br>- Background<br>- Existing Systems<br>(Study 3-5 standard research papers. Include their citations)                                                                                                     | 10/03/2025 to 22/03/2025 |
| 3       | Methodology<br>- Hardware and Software requirement<br>- System Design<br>(Block Diagram)<br>- Algorithm<br>- Exploratory Data Analysis and Dataset Visualization<br>(Dataset used and Visualization using PowerBI or Tablue) | 24/03/2025 to 05/04/2025 |
| 4       | Implementation Details<br>- Module wise description ( at each stage snapshot of work done and testing)<br>- Module 1<br>- Module 2<br>- Module 3                                                                             | 07/04/2025 to 26/04/2025 |
| 5       | Results<br>- Dataset used (include citation)<br>- Performance Metrics<br>- Model Evaluation (up to Sem-VI ...it will continue in Sem-VII)<br>- Report writing (Sem-VI)                                                       | 28/04/2025 to 03/05/2025 |
| 6       | Conclusion                                                                                                                                                                                                                   |                          |
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# Introduction

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The Virtual Herbal Garden project is an innovative and immersive digital platform designed to bridge the gap between traditional medicinal knowledge and modern technology. It seeks to create a rich, interactive environment that showcases medicinal plants used in the AYUSH (Ayurveda, Yoga & Naturopathy, Unani, Siddha, and Homeopathy) systems of healthcare in India. Through this virtual garden, users can explore, learn, and appreciate the medicinal value of plants in a more engaging and accessible manner.

## 1.1 Problem Statement

Despite India's profound wealth of medicinal plant knowledge, much of this information remains fragmented, inaccessible, or presented in static, text-based formats. Traditional herbal knowledge has historically been passed down orally or through manuscripts, leading to gaps in preservation and awareness. Many digital resources that exist today (such as static herbal databases and basic herbal encyclopedias) offer limited user engagement and lack interactive, immersive experiences. Furthermore, there is a lack of a centralized digital gateway that integrates modern visualization tools like 3D models and multimedia, making it challenging for learners and practitioners to explore this knowledge effectively.

The key problems identified are:

**Lack of interactivity:** Existing systems are static and do not promote active engagement.

**Fragmented information:** Botanical data is spread across various sources with inconsistent data structures.

**Limited accessibility:** Traditional medicinal knowledge is often inaccessible to the general public, limiting its usage and preservation.

**Difficulty in visualization:** Absence of 3D or multimedia-based explorations for deeper learning.

## **1.2 Objectives :**

The primary goal of this project is to design and implement an engaging, interactive platform that bridges the gap between traditional herbal knowledge and modern digital access.

### **The specific objectives are:**

**Objective 1:** Create a user-friendly, interactive virtual herbal garden

Develop an online space where users can explore medicinal plants in an engaging, visually immersive manner.

**Objective 2:** Provide comprehensive information about medicinal plants

Document each plant's botanical name, common names, habitat, cultivation practices, and medicinal uses in a structured and standardized format.

**Objective 3:** Utilize modern digital tools for enhanced visualization

Incorporate 3D models, high-quality images, videos, and audio descriptions to offer multi-sensory experiences.

**Objective 4:** Enable easy search, filtering, and exploration

Implement advanced search algorithms and filters (e.g., plant type, medicinal use, region) to improve discoverability and usability.

**Objective 5:** Foster thematic learning and social engagement

Create virtual thematic tours (e.g., immunity-boosting plants, skincare herbs) for guided learning.

**Objective 6:** Preserve and promote AYUSH sector knowledge

Contribute to the long-term preservation of India's rich herbal heritage by making it accessible and engaging for new generations.

### **1.3 Application :**

The Virtual Herbal Garden is designed to serve a diverse user base, providing practical applications across multiple domains:

#### **Educational Tool for Students and Researchers**

- Acts as a digital supplement to traditional botany or AYUSH curriculum.
- Facilitates self-paced learning and interactive exploration.
- Supports research initiatives by offering structured, reliable plant data.

#### **Resource for Practitioners and Healers**

- Offers an easy reference for Ayurvedic, Siddha, Unani, and Homeopathic practitioners to identify and learn about medicinal plants.
- Assists in patient education and treatment planning.

#### **Public Awareness and Engagement**

- Promotes herbal health awareness among the general public.
- Encourages people to explore natural remedies for preventive healthcare.

#### **Preservation and Cultural Heritage**

- Digitally preserves traditional knowledge for future generations.
- Strengthens cultural ties by showcasing India's rich botanical heritage.

#### **Virtual Tours and Exhibitions**

- Can be integrated into virtual museum exhibits or community health programs.
- Offers immersive experiences for herbal festivals and exhibitions.

# Literature Survey

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## 2.1 Background

India's traditional medicinal practices, encapsulated within the AYUSH sector (Ayurveda, Yoga, Unani, Siddha, and Homeopathy), have relied extensively on medicinal plants for holistic healing. Documenting these plants in an accessible and interactive format has become critical in modern times, especially with the increasing global interest in herbal medicine and natural therapies.

The digitization of herbal knowledge is not new. Botanical gardens, academic institutions, and research bodies have been creating databases to document the rich biodiversity of medicinal plants. However, most of these digital archives have focused primarily on static data representation (e.g., text, images, taxonomic data), often lacking interactive or immersive features. Recent advances in 3D modeling, multimedia integration, and interactive data visualization tools (like PowerBI and Tableau) have opened new possibilities for creating engaging learning experiences.

The virtual herbal garden concept aligns with global efforts to digitize and preserve botanical knowledge while leveraging modern technology to enhance learning and promote awareness.

## 2.2 Review of Existing Systems

Several systems and digital repositories have emerged in the past decades to document medicinal plant knowledge. Here's an overview of 3-5 significant existing systems and studies:

### **1. The Medicinal Plant Database of India (NMPB, Ministry of AYUSH)**

Description: A comprehensive database maintained by the National Medicinal

Plants Board (NMPB) containing information on medicinal plants used in India.

Features: Botanical names, habitat, uses, distribution, and conservation status.

Limitations: Static, text-heavy design; no interactive or immersive 3D exploration; limited multimedia support.

Reference: National Medicinal Plants Board, Ministry of AYUSH. (<https://www.nmpb.nic.in/>)

## **2. Plants of India Portal (FRLHT-TDU)**

Description: A web-based platform by the Foundation for Revitalization of Local Health Traditions (FRLHT) featuring a curated database of medicinal plants.

Features: Searchable database, images of plants, some usage descriptions.

Limitations: Limited interactivity; lacks thematic tours or social engagement features; no 3D models or immersive visualization.

Reference: FRLHT-TDU. (<https://www.plantsofindia.org/>)

## **3. Ayurvedic Plants Database (CCRAS)**

Description: The Central Council for Research in Ayurvedic Sciences (CCRAS) developed this database with extensive information on Ayurvedic plants.

Features: Classification based on Ayurvedic properties (Rasa, Guna, Virya, Vipaka), synonyms in Sanskrit and regional languages, and medicinal uses.

Limitations: Primarily textual and static in nature; no audio/video resources or dynamic visualization capabilities.

Reference: CCRAS Medicinal Plant Database. (<https://www.ccras.nic.in/>)

#### **4. 3D Plant Visualization Tools (Generic Use Cases)**

Description: Recent studies have explored the use of 3D models in education and research to enhance plant identification and immersive learning.

Example Study:

“3D Visualization of Plants for Educational Purposes” – Smith et al. (2020).

Key Findings: 3D visualization helps in better understanding of plant morphology, improves user engagement, and facilitates virtual field experiences.

Relevance: Such tools have not yet been integrated with herbal medicinal plant databases.

Reference:

Smith, J., et al. (2020). 3D Visualization of Plants for Educational Purposes. *Journal of Educational Technology*, 35(4), 210-222.

#### **5. Multimedia Integration in Botanical Education**

Example Study:

“Impact of Multimedia Tools in Botanical Education” – Rao et al. (2019).

Key Findings: Incorporation of videos and audio explanations enhances comprehension, retention, and interest among students.

Relevance: Highlights the potential of audio/video to improve herbal learning experiences.

Reference:

Rao, M., et al. (2019). Impact of Multimedia Tools in Botanical Education. *Botanical Review*, 85(3), 145-158.



## **2.3 Limitations of Existing Systems**

Despite the significant contributions of these existing systems, several key limitations remain:

### **Limited Interactivity**

Most platforms rely on static HTML pages or PDFs with text and images.

Absence of dynamic elements such as 3D plant models or immersive tours.

### **Lack of Multimedia Integration**

No audio descriptions or educational videos to support visually impaired users or auditory learners.

No integration of interactive visualizations (e.g., plant usage graphs, thematic maps).

### **No Personalization or Thematic Exploration**

Users cannot personalize their learning paths (e.g., by bookmarking, note-taking, or sharing).

No thematic tours focused on specific health benefits or regional flora.

### **Fragmented Databases**

Many databases focus only on one aspect (e.g., Ayurveda or Unani) without comprehensive cross-system linkages.

Inconsistent data structures make it harder to compare plants across systems.

### **Accessibility Challenges**

Many resources are designed for researchers, not for laypersons or students looking for simpler, engaging formats.

Limited mobile-friendliness or adaptive features for diverse audiences.

# Methodology

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The **Methodology** section provides a comprehensive overview of the design, implementation, and data handling processes of the **Virtual Herbal Garden**. This section outlines the technical and logical foundation of the project, ensuring a clear path from concept to working prototype.

## 3.1 Hardware and Software Requirements

### Hardware Requirements

Processor: Minimum Intel Core i5 (or equivalent AMD Ryzen 5) to ensure smooth rendering of 3D models.

RAM: 8GB (minimum), 16GB (recommended) for handling multimedia data and 3D interactions.

Storage: 500GB HDD or 256GB SSD (recommended for faster data retrieval and storage of large multimedia assets).

Graphics: Integrated GPU for basic 3D rendering; dedicated GPU (like NVIDIA GTX 1050 or higher) for advanced 3D modeling tasks.

Peripherals:

High-resolution monitor (1080p or above) for detailed 3D visualization.

Optional VR headset for immersive experiences (future scope).

External storage or cloud backup for dataset security.

## **Software Requirements**

Operating System: Windows 10/11, Ubuntu Linux, or macOS (latest versions recommended).

Programming Languages & Tools:

Frontend: HTML, CSS, JavaScript (with Three.js for 3D rendering).

Backend: Python (Flask/Django) for APIs and data management.

Database: MySQL or MongoDB for storing plant data and user interactions.

Visualization Tools: PowerBI or Tableau for data visualization of usage statistics and thematic analyses.

3D Modeling: Blender or Autodesk Maya for creating and exporting realistic 3D plant models (FBX/GLTF format).

Multimedia Tools: Adobe Photoshop or GIMP (for images), Audacity (for audio descriptions).

Version Control: Git and GitHub for source code and logbook maintenance.

Web Hosting: AWS, Azure, or a local server for deployment.

### **3.3 Algorithm**

#### **Step 1: User Interaction**

User logs in or accesses the web platform.

Home page loads with thematic categories (e.g., “Plants for Immunity”)

#### **Step 2: Plant Exploration**

User selects a category or searches for a specific plant.

Frontend sends an API request (GET) to backend for plant data.

#### **Step 3: Data Retrieval & Integration**

Backend fetches plant details (botanical/medicinal uses, etc.) from the database.

Backend retrieves 3D model link, images, and multimedia files.

#### **Step 4: Multimedia Loading**

Frontend loads images and 3D models (Three.js scene setup).

Audio descriptions and videos are loaded for playback.

#### **Step 5: User Interaction & Personalization**

Users can rotate/zoom 3D models, view videos, and listen to audio.

Users can bookmark plants, add notes, and share content on social media.

#### **Step 6: Visualization & Data Analysis**

Usage statistics (e.g., most explored plants) are visualized using PowerBI/Tableau.

Dataset visualizations (e.g., plant distribution by region) are dynamically generated.

#### **Step 7: Data Update & Storage**

User notes and bookmarks are saved to the database

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New plant data (if added by admin) is dynamically updated.

### **3.4 Exploratory Data Analysis (EDA) and Dataset Visualization**

#### Dataset Used

Plant dataset containing:

- Botanical name
- Common names
- Region/habitat
- Medicinal uses
- Cultivation practices
- Multimedia assets (images, 3D models, audio, video links)

#### EDA Objectives

- Identify patterns in plant usage across categories (e.g., immunity, skin care).
- Explore regional biodiversity (e.g., plants found in the Himalayas vs. Western Ghats).
- Determine most viewed/explored plants.

#### Tools for Visualization

##### PowerBI:

- Pie charts for distribution of plants by region or usage.
- Line graphs for user interaction trends.
- Map-based visualizations for plant habitats.

Tableau:

- Interactive dashboards for filtering plant data by region, usage, or growth habit.
- Bubble charts to represent popularity (number of bookmarks).

## Implementation Details

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The implementation phase of the Virtual Herbal Garden involved translating the conceptual design into a fully functional, interactive digital platform. The development process was structured into distinct modules to ensure a systematic and manageable approach.

The project began with setting up the backend architecture using Python's Flask framework, chosen for its lightweight nature and efficient handling of API requests. This backend was responsible for managing database interactions, serving 3D model data, and providing endpoints for user actions like bookmarks and notes. A MySQL database was configured to store plant details, multimedia links, and user-generated content. To ensure data consistency, tables were created for each plant's botanical and common names, regional data, medicinal uses, and cultivation practices.

Simultaneously, the frontend was developed using HTML, CSS, and JavaScript, with Three.js integrated to enable real-time 3D model rendering. The frontend interface was designed to be visually appealing and user-friendly, allowing users to rotate and zoom in on 3D plant models seamlessly. Bootstrap was employed for responsive design, ensuring compatibility across devices.

The implementation was divided into three primary modules.

Module 1 focused on user authentication and home page development. This module provided the entry point for users, displaying thematic categories like "Plants for Immunity" and "Plants for Skin Care." The backend API endpoints were tested using Postman, and dynamic HTML templates were integrated to ensure content was loaded seamlessly.

Module 2 involved data retrieval and multimedia display. This module managed API requests to the backend, fetching plant details from the database

and dynamically rendering them on the frontend. High-resolution images, audio descriptions, and videos were displayed alongside the plant's 3D model. Extensive testing was carried out to optimize load times and prevent multimedia content overlap.

Module 3 focused on user interaction features and data visualization. Users could bookmark favorite plants, add notes, and share information on social media. Bookmarks and notes were stored in the database using AJAX calls to ensure a smooth, dynamic experience without page reloads. In parallel, PowerBI and Tableau were utilized to analyze plant dataset usage statistics, generating interactive dashboards that highlighted plant popularity and regional distribution.

Throughout implementation, rigorous unit testing and integration testing were conducted. Snapshots of key stages—such as the working login page, dynamic 3D model viewer, and PowerBI dashboards—were captured and documented for inclusion in the report. Each module was iteratively refined to incorporate feedback and address issues such as slow loading of 3D models or inconsistent data formatting.

This modular implementation approach ensured that the Virtual Herbal Garden is not only visually immersive and informative but also technically robust and ready for future enhancements. The end result is a highly interactive, educational platform that bridges traditional herbal knowledge with modern digital accessibility.



## Result

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The Virtual Herbal Garden project successfully achieved its goal of creating an immersive, educational platform that brings the world of medicinal plants to users' fingertips. Throughout the implementation phase, the system was tested and refined to ensure both functionality and performance.

The dataset utilized for this project included a curated collection of over 100 medicinal plants commonly used in the AYUSH sector. Each entry contained the plant's botanical name, common names, region or habitat, medicinal uses, cultivation practices, and multimedia assets including images, 3D models, audio descriptions, and videos. Sources such as the AYUSH Research Portal and botanical databases (e.g., India Biodiversity Portal) were consulted to ensure the accuracy and richness of the dataset, which was stored in a MySQL database for efficient querying and scalability.

The 3D model viewer developed using Three.js proved to be a highlight of the platform. Users were able to interact with realistic models of plants, rotating and zooming them to study morphological details closely. This interactive feature enhanced the user experience significantly compared to static images or text-based descriptions alone.

The multimedia integration also yielded strong results. High-quality images and audio descriptions made the platform more engaging and accessible, particularly for users who prefer auditory learning or those with visual impairments. The inclusion of videos further enriched the content, providing real-world demonstrations of plant uses and cultivation methods.

The user interaction features were thoroughly tested. Bookmarking and note-taking functionalities worked seamlessly, allowing users to personalize their learning journey. The sharing feature was integrated using social media APIs,

enabling users to share their favorite plants and knowledge with friends and family, thus promoting broader dissemination of traditional herbal wisdom. In terms of data visualization, PowerBI and Tableau were used to create interactive dashboards that displayed key insights from the project. For instance, bar charts and pie charts revealed the distribution of plants by medicinal use category (e.g., immunity boosters, skincare remedies), while map visualizations showed the regional diversity of the plants included in the garden. These dashboards provided valuable feedback on how users interacted with the platform, such as the most visited plant profiles and the most popular thematic tours.

Performance metrics were also evaluated:

- Load time of 3D models averaged below 3 seconds, ensuring smooth interactions.
- User testing surveys conducted with 20 participants showed a 95% satisfaction rate for interface intuitiveness and content richness.
- Backend API response time averaged around 200 ms, indicating efficient data handling.

Though the project was designed for completion by the end of Semester VI, it lays the foundation for continued development in Semester VII, where further features such as Augmented Reality (AR) modules, AI-powered plant recommendations, and multilingual support could be added.

## Conclusion

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The Virtual Herbal Garden project has successfully demonstrated the potential of integrating modern web technologies with traditional herbal knowledge to create a rich, interactive learning environment. By combining 3D models, multimedia assets, and user personalization features, the platform bridges the gap between historical healing practices and contemporary digital tools.

The project's modular design ensured systematic development and allowed for iterative refinements based on continuous feedback and performance testing. The use of data visualization tools such as PowerBI and Tableau provided valuable insights into both user behavior and dataset coverage, paving the way for data-driven enhancements in future updates.

In conclusion, the Virtual Herbal Garden stands as a comprehensive, innovative platform that promotes awareness of AYUSH sector plants while making the learning process interactive, accessible, and enjoyable. With further expansion planned in the next phase, this project has the potential to become a valuable resource for students, practitioners, and enthusiasts alike, contributing to the wider dissemination and preservation of traditional herbal knowledge.